PRATIK NANDAPURKAR

QUALITY ASSURANCE ENGINEER (DATA SCIENCE) - EPIQ SYSTEMS

+918668659287

pratik.nandapurkar17@gmail.com

Hyderabad, Telangana

EDUCATION

2016 - 2020
Bachelor of Engineering
University of Pune , CGPA- 8.69

2015
HSC
State Board of Maharashtra - 79.54%

2013
SSC
State Board of Maharashtra - 89%

SKILLS

1 . Python

- Python libraries like Numpy , Pandas , Scikit Learn , Matplotlib , Scipy , Math , Seaborn
- Regular Expression , functional programming, python data types , anonymous function, date time
- Web scrappig, Pytesseract, Image2Pdf ,Pdf2Image
- VS Code , Jupyter Notebook
- Object oriented programming, class , object, Inheritance , Encapsulation , Polymorphism , Abstraction
- Iterator , generator , decorator

2. Machine Learning

- **Supervised and unsupervised ML Algorithms** such as Linear Regression , Gradient Descent , Ridge & Lasso Regression, Logistic Regression , Naïve Bayes , MultinomialNB , GaussianNB , K Nearest Neighbor , Support Vector Machine , Decision Tree, Ensemble Learning such as Random Forest, Ada-Boost and Voting , Stacking Gradient Boosting, XGBoost, K-means Clustering, Hierarchical Clustering.

About me

Currently working as a Quality Assurance Engineer(Data Science) and an aspiring AI Engineer with 5+ years of experience in Data Science and Generative AI including profound experience in transforming business requirement into analytical model, working on Generative AI models as well as end to end testing, designing algorithms and strategic solutions that scales across massive volumes of data.

Work Experience

Epiq Systems | Quality Assurance Engineer - Data Science Hyderabad

Sept 2023 - Present

- Working as a key member in various stages of project and responsible for having SPOC to stakeholders with ease and professionalism.
- Ability to create custom models using Gemini, Huggingface & OpenAI's API.
- Thorough understanding of Generative AI and Natural Language Processing.
- Involved in end to end testing(backend and frontend) on created models and software applications.
- Knowledge of model training, fine-tuning and Prompt Engineering.

Tech Mahindra | Machine Learning Engineer May 2020 - Aug. 2023 Pune

- Worked as a Machine Learning Engineer with experience in Python, Data Science and Machine learning with expertise in Finance & Banking as well as NLP project.
- Thorough understanding of Probability and Statistics, Bayesian methods, .
- Experience in data management tools - NoSQL and SQL Databases.
- Knowledge of Python's Data Cleaning, Data Analysis and different Machine Learning libraries.
- Source code management and Version control system using Git and GitHub.

Projects

1. Privilege Detection - EPIQ SYSTEMS

Type- NLP Gen-AI

- Creating a dataset from multiple schemas and cleaning the data.
- Creating and comparing Generative AI models using different frameworks (langchain, llamaindex, aws-bedrock, Jarvis and mesh tensorflow)
- Fine tuning a model based on particular data.
- Creating testcases and validation for user story.
- End to end evaluating created/trained model.
- Performing load testing and UI testing on deployed application.
- Data Wrangling, Preprocessing, Embeddings and Encoding.

3. Natural Language Processing

- NLTK
- EDA
- Data Wrangling and Preprocessing
- Tokenization
- Word Embedding Methods
- Information Extraction
- GoogleTrans , LangDetect

4. SQL and NoSQL

- MySQL
- PostgreSQL

5. Generative AI

- ANN, CNN
- LANGUAGE TRANSLATION
- SEQUENCE TO SEQUENCE MAPPING USING RNN, LSTM, GRU, ENCODER, DECODER ,ATTENTION, SELF ATTENTION (TRANSFORMERS).
- FINE TUNING
- RETRIEVAL AUGMENTED GENERATION
- LANGCHAIN , LLAMAINDEX

6. Git and Github

7. Cloud Service - Amazon AWS

- S3
- Sagemaker
- EC2

8. Cloud Service - Azure

- Azure Storage Explorer
- Azure API Management
- Azure Function apps
- Resource groups
- Container apps and Microservices
- Developers Portal
- Azure Virtual Machine & ML Studio
- Azure Devops

9. Prompt Engineering

10. Software Testing

- Tools like Postman API, Jmeter, python scripts and Swagger image.
- UI Testing
- Backend Testing.
- Various testing techniques like load testing, smoke testing, regression testing etc.

2. Digital Assistant using Retrieval Augmented Generation - EPIQ SYSTEMS

Type- NLP Gen-AI RAG

- Creating a dataset from multiple schemas and cleaning the data.
- Setting up the system prompts for chatbot.
- Data Wrangling, Preprocessing, Embeddings and Encoding.
- Creating question-answering data by implementing an augmented toolkit for fine tuning the model/chatbot.
- Fine tuning model.
- Testing the RAG chatbot using pre-defined testcases.
- Creating testcases for various features implemented over the chatbot UI.
- End to end evaluating created/trained model.
- Performing load testing using Jmeter, python scripts and UI testing on deployed application.
- Train the developed model and run evaluation experiments.

3. EPIQ Discovery Management Console - EPIQ SYSTEMS

- Involved in various stages of project including POC
- Management console has various applications such as Gen AI Text Summarization, Sentiment analyzer, Image Narrator, Prompt Management, Document management.
- Text Summarizer is built on the top of the customized model from OpenAI's GPT 3.5 Turbo and GPT 4.0 mini.
- Involved in backend testing of Sync and Async Gen AI Text Summarization.
- Worked on priority queues and reprioritization(backend) for Async Text Summarization.
- Involved in end to end model building on sentiment analyzer and Image Narrator.
- Implemented Estimated Time Predictor for Text Summarization and Sentiment Analyzer.
- Sentiment Analyzer : Preprocessing techniques such as Contraction Mapping , Punctuation , Stopwords Removal , LangDetect , Translate, sent_tokenize , word_tokenize , WhitespaceTokenizer , normalization, accented characters handling.
- Creating testcases for various features implemented in the UI
- End to end evaluating created/trained model.
- Performing load testing using Jmeter, python scripts and UI testing on deployed application.
- Backend testing using Postman and Swagger Image.

4. Loan Status Prediction Management System - Tech Mahindra

Type- Finance and Banking

- Develop advanced programs to extract the data needed, prepare data for further analysis.
- Work on all stages/sprints of a data science / ML project - exploration and conceptualization, POC (proof of concept), data preparation, model development and testing, deployment, monitoring and debugging, continuous improvement
- Partner with product, data engineering and ML engineering teams to ensure seamless productionisation of developed algorithm and management of the full model cycle.

- Creating Testcases and logging the same in Azure.
- Creating bug and re-testing after resolution.
- Pre and Post model deployment testing on various environment like dev, qa, uat and production.

11. VECTOR DATABASES

- MongoDB, ChromaDB
- Cassendra DB, Pinecone
- Opensearch, LanceDB

12. OpenAI ChatGPT, HuggingFace

13. Statistics and Data Visualisation

14 FLASK

15. Project Management Tool

- Jeera
- Azure Dashboard

SOFT SKILLS

- Keen learner
- Team player
- Adaptability
- Organized efficient
- Detail-oriented
- Creative thinking
- Problem Solving
- Multi_tasking

PERSONAL INFORMATION

- Birthdate - 27th June 1997
- Gender - Male
- Nationality - Indian
- Languages Known - English, Hindi, Marathi
- Certification - English communication nobel communications , Pune
- Hobbies - reading books , singing , sketching, travelling , listening songs, writing diary.

- Univariate analysis , Bivariate analysis , handling missing values and outliers , feature scaling methods like Normalization , Standardization , feature selection techniques , data splitting into train and test, model optimization using hyper parameter tunning , Ridge Regression , Lasso Regression , handling imbalance data , feature encoding methods(Label Encoding , One Hot Encoding) , data cleaning.

5. Product Review Sentiment Analysis - Tech Mahindra

Type - NLP

- Design and develop natural language processing systems. Define appropriate datasets for language learning.
- Use effective text representations to transform natural language into useful features.
- Develop NLP systems according to requirements.
- Train the developed model and run evaluation experiments.
- Find and implement the right algorithms and tools for NLP tasks. Perform statistical analysis of results and refine models constantly keep up to date within the field of machine learning. Maintain NLP libraries and frameworks.
- Implement changes as needed and analyze bugs.
- IN EDA , Key Phrase Extraction techniques such as RAKE , YAKE , Ngrams such as Unigram , Bigrams , Trigrams and WordCloud
- Stemming and Lemmatization techniques such as LancasterStemmer , PorterStemmer ,WordNetLemmatizer, SnowballStemmer RegexStemmer , XeroxStemmer .
- Autocorrection libraries such as Autocorrect , Textblob , Symspell
- Preprocessing techniques such as Contraction Mapping , Punctuation , Stopwords Removal , LangDetect , Translate, sent_tokenize , word_tokenize , WhitespaceTokenizer , normalization, accented characters handling.
- Word embedding technique such as CountVectorizer , TfidfVectorizer , Word2Vec (CBOW , Skipgram) , Doc2vec , Fast-text.

6. Vehicle's Physical Damage Insurance Fraud Detection - Tech Mahindra

Type- Finance and Banking

- Design, develop and implement analytical solutions using a variety of commercial and open-source tools (common tools include Python).
- Develop and embed automated processes for predictive model validation, deployment, and implementation.
- Ensure Analytics model quality.
- Connect and collaborate with subject matter experts in R&D.
- Make impactful contributions to internal discussions on emerging machine learning methodologies.
- Educate the organization both from IT and the business perspectives on these new approaches, such as testing hypotheses and statistical validation of results.
- Work closely with the Engineering and Tech team to convert POC into a fully scalable product.
- Collaboratively manage projects from timeline creation to project completion, managing expectations with leaders and colleagues.
- EDA , Preprocessing and feature engineering techniques. Univariate and multivariate analysis of features. Detecting, handling , imputing outliers and missing values.